Here is **Everything You Need to Know About Sets in Python** — clear and complete.

Python Set — Complete Guide

1. What is a Set in Python?

A **Set** is:

- Unordered collection
- No duplicate values
- Mutable (can change after creation)
- Written using `{}` braces

Example:

```python
my_set = {1, 2, 3, 4}
print(my_set)
```

Output:

```
{1, 2, 3, 4}
```

2. Characteristics of Sets

- No duplicates
- Unordered
- Mutable
- No indexing
- Fast membership testing

Example:

```python
s = {1, 2, 2, 3, 3}
print(s)
```

Output:

```
{1, 2, 3}
```

Duplicates automatically removed.

3. Creating Sets

Method 1: Using {}

```python
a = {1, 2, 3}
```

Method 2: Using set()

```python
a = set([1, 2, 3])
```

Empty Set (Important)

```python
a = set()   # correct
```

Wrong:

```python
a = {}  # This creates dictionary
```

4. Set Data Types

Set can store:

- Integers
- Strings
- Floats
- Mixed values

Example:

```python
s = {1, "hello", 3.5}
print(s)
```

5. Adding Elements

add()

```python
s = {1, 2, 3}
s.add(4)
print(s)
```

Output:

```
{1, 2, 3, 4}
```

6. Adding Multiple Elements

update()

```python
s = {1, 2}
s.update([3, 4, 5])
print(s)
```

Output:

```
{1, 2, 3, 4, 5}
```

7. Removing Elements

remove()

```python
s = {1, 2, 3}
s.remove(2)
print(s)
```

If element not found → Error

discard()

```python
s.discard(5)
```

No error if element not found.

pop()

Removes random element

```python
s = {1, 2, 3}
s.pop()
print(s)
```

clear()

Remove all elements

```python
s.clear()
print(s)
```

Output:

```
set()
```

8. Set Operations

1. Union

Combines two sets

```python
a = {1, 2, 3}
b = {3, 4, 5}

print(a | b)
```

OR

```python
print(a.union(b))
```

Output:

```
{1, 2, 3, 4, 5}
```

2. Intersection

Common elements

```python
print(a & b)
```

OR

```python
print(a.intersection(b))
```

Output:

```
{3}
```

3. Difference

Elements in first set but not second

```python
print(a - b)
```

OR

```python
print(a.difference(b))
```

Output:

```
{1, 2}
```

4. Symmetric Difference

Elements not common

```python
print(a ^ b)
```

OR

```python
print(a.symmetric_difference(b))
```

Output:

```
{1, 2, 4, 5}
```

9. Set Comparison

subset

```python
a = {1, 2}
b = {1, 2, 3}

print(a.issubset(b))
```

Output:

```
True
```

superset

```python
print(b.issuperset(a))
```

Output:

```
True
```

disjoint

No common elements

```python
a = {1, 2}
b = {3, 4}

print(a.isdisjoint(b))
```

Output:

```
True
```

10. Loop Through Set

```python
s = {1, 2, 3}

for i in s:
    print(i)
```

11. Length of Set

```python
s = {1, 2, 3}
print(len(s))
```

Output:

```
3
```

12. Check Element Exists

```python
s = {1, 2, 3}

print(2 in s)
```

Output:

```
True
```

13. Copy Set

```python
a = {1, 2, 3}
b = a.copy()

print(b)
```

14. Frozen Set (Immutable Set)

Cannot change after creation

```python
fs = frozenset([1, 2, 3])
print(fs)
```

You cannot use:

```python
fs.add(4)  # Error
```

15. Set Methods Summary

Method	Description
add()	Add element
update()	Add multiple elements
remove()	Remove element
discard()	Remove without error
pop()	Remove random element
clear()	Remove all

Method	Description
union()	Combine sets
intersection()	Common elements
difference()	Subtract sets
symmetric_difference()	Not common
issubset()	Check subset
issuperset()	Check superset
isdisjoint()	No common elements
copy()	Copy set

16. Real Example

Python

```python
students_A = {"Ram", "Hari", "Sita"}
students_B = {"Sita", "Gita", "Ram"}

print("Union:", students_A | students_B)
print("Intersection:", students_A & students_B)
print("Difference:", students_A - students_B)
```

17. When to Use Sets

Use sets when:

- Remove duplicates
- Fast searching
- Mathematical operations
- Unique values needed

Example:

Python

```python
numbers = [1, 2, 2, 3, 4, 4]
unique = set(numbers)

print(unique)
```

Output:

```
{1, 2, 3, 4}
```

If you'd like, I can also give:

- Practice questions
- Interview questions
- Cheat sheet (1 page)
- Real project examples

Just tell me 👍